A Finite Schur–Feshbach Test for Reflected Wilson Blocks: An Explicit Four-Dimensional $SU(2)$ Heat-Kernel Lattice Realization

Andre Freitas

Affiliation and contact information to be supplied

July 10, 2026

Abstract

We give a fully specified finite-cutoff realization of a Schur–Feshbach positivity test for reflected Wilson observables. The matrix statement is the rank-one specialization of standard second-order geometry of the positive-semidefinite cone: if

$$q_\varepsilon = \mathbf{1}\mathbf{1}^* + \varepsilon H + \varepsilon^2 K + o(\varepsilon^2),$$

then, after eliminating the rank-one direction, the first two coefficients are

$$A = PHP, \qquad C = P\big(K - n^{-1} H e e^* H\big)P, \qquad e = n^{-1/2}\mathbf{1}, \quad P = I - ee^*.$$

The field-theoretic contribution is an explicit same-theory block. On a finite four-dimensional reflection-symmetric lattice with gauge group $SU(2)$ and heat-kernel plaquette weight, we define three closed hairpin loops sharing a base path. A spatial-reflection odd Wilson mode X and an even centered Wilson mode Y generate three positive-time observables

$$F_i^\varepsilon = 1 + \sqrt{\varepsilon}\, b_i X + \varepsilon c_i Y, \qquad b = (1,-1,0)^T, \quad c = (0,0,1)^T.$$

Their reflected Gram matrix is exactly

$$N_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon\sigma b b^T + \varepsilon^2 \tau c c^T, \qquad \sigma = \langle \Theta X, X\rangle_{\mathrm{OS}} \geq 0, \quad \tau = \langle \Theta Y, Y\rangle_{\mathrm{OS}} \geq 0.$$

Diagonal correlation normalization is realized by rescaling the same observables, hence is an exact congruence rather than an uncontrolled entrywise quotient. The normalized block has a rank-one limit and yields

$$PHP = \sigma b b^T, \qquad C = -\sigma^2 b b^T + \frac{\tau}{9} z z^T, \qquad z = (1,1,-2)^T.$$

Thus the first reduced eigenvalue is 2σ, while the corrected second-order null-sector eigenvalue is $2\tau/3$. The uncorrected quantity contains a spurious σ^2 term, which the Schur–Feshbach correction removes. All statements up to this point are exact at finite cutoff. Geometric collar collapse, uniform nondegeneracy along a continuum trajectory, endpoint cumulant estimates, infinite volume, and a spectral-gap implication are isolated as explicit open modules rather than claimed results.

Contents

1 Scope, novelty, and claim boundary **2**

2 Finite reflected Gram matrices and the Schur–Feshbach test **3**

3 A reflection-positive four-dimensional lattice model **4**
3.1 Finite lattice, action, and reflections 4
3.2 Three explicit closed hairpins 5

4 The exact three-hairpin source block **6**
4.1 Positive-time observables and exact Gram matrix 6
4.2 Same-theory correlation normalization 7

5 Coefficient extraction and the finite certificate **8**

6 Finite evaluation and reproducibility protocol **10**

7 Open modules toward a geometric collar and a mass-gap programme **10**
7.1 Cutoff trajectory and geometric collision 11
7.2 Correct endpoint/null-channel bookkeeping 13
7.3 From OS positivity to a spectral gap 15

8 Conclusion **16**

A Normalization algebra **16**

B Exact block algebra and uniform remainder **17**

C Five-point fourth-derivative certificate **18**

D Checklist for the next manuscript revision **18**

1 Scope, novelty, and claim boundary

The Yang–Mills existence and mass-gap problem asks for a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group and a positive mass gap in the same reconstructed theory [10, 11]. Osterwalder–Schrader reconstruction provides a Euclidean route, in which reflection positivity is one indispensable axiom [2, 3]. At finite lattice spacing, standard plaquette gauge actions furnish reflection-positive regulated theories [6, 5].

This paper makes a deliberately narrower contribution. It does not construct a continuum Yang–Mills measure, establish all OS axioms, prove exponential clustering, or prove a mass gap. It does two finite things:

(i) it states the rank-one Schur–Feshbach coefficient test for a singular PSD matrix family; and

(ii) it implements that test on an explicit four-dimensional $SU(2)$ lattice Wilson block whose action, reflection, observables, normalization, rank-one limit, coefficients, and finite margins are all specified.

The Schur-complement curvature term is not presented as a new discovery in matrix analysis. It is a rank-one instance of the established second-order geometry of semidefinite constraints [8, 9]. The purpose of retaining the finite theorem is to make the Wilson calculation self-contained and to show exactly where the curvature correction enters. The substantive advance over a purely schematic Wilson/OS discussion is the same-theory realization in Sections 3–5.

Claim boundary. All results through Corollary 5.4 hold on a fixed finite lattice, and the source-collision expansion is uniform over arbitrary cutoff families because the primitive Wilson observables are bounded. Strict positive margins require the finite nondegeneracy conditions $\sigma > 0$ and $\tau > 0$. No uniform lower bound on these quantities as the cutoff is removed is proved here.

2 Finite reflected Gram matrices and the Schur–Feshbach test

Let Θ be an antilinear Euclidean reflection and let $\mathcal{A}_+$ be an algebra of positive-time observables. Reflection positivity means

$$\sum_{i,j=1}^{n} \overline{c_i} c_j \, \langle (\Theta F_i)^* F_j \rangle \geq 0 \tag{1}$$

for every finite family $F_1, \ldots, F_n \in \mathcal{A}_+$ and every $c \in \mathbb{C}^n$. Equivalently, the reflected Gram matrix

$$N_{ij} = \langle (\Theta F_i)^* F_j \rangle \tag{2}$$

is positive semidefinite.

Suppose a normalized finite block has a rank-one collision expansion

$$q_\varepsilon = \mathbf{1}\mathbf{1}^* + \varepsilon H + \varepsilon^2 K + R_\varepsilon, \qquad \|R_\varepsilon\| = o(\varepsilon^2), \tag{3}$$

where $H, K \in \mathrm{Herm}_n(\mathbb{C})$. Define

$$e = n^{-1/2}\mathbf{1}, \qquad P = I - ee^*. \tag{4}$$

Then P projects onto $\mathbf{1}^\perp$ and $\mathbf{1}\mathbf{1}^* = nee^*$.

Theorem 2.1 (Rank-one Schur–Feshbach certificate)**.** *Let q_ε satisfy (3). Set*

$$\lambda_\varepsilon = e^* q_\varepsilon e, \qquad \mathcal{F}_\varepsilon = P q_\varepsilon P - P q_\varepsilon e \, \lambda_\varepsilon^{-1} e^* q_\varepsilon P. \tag{5}$$

For all sufficiently small $\varepsilon > 0$, $\lambda_\varepsilon > 0$ and

$$q_\varepsilon \succeq 0 \quad \Longleftrightarrow \quad \mathcal{F}_\varepsilon \succeq 0. \tag{6}$$

Moreover,

$$\mathcal{F}_\varepsilon = \varepsilon A + \varepsilon^2 C + o(\varepsilon^2), \qquad A = PHP, \qquad C = P\big(K - n^{-1} H e e^* H\big)P. \tag{7}$$

If $q_\varepsilon \succeq 0$ for all sufficiently small $\varepsilon > 0$, then

$$A \succeq 0 \quad \text{on } \mathbf{1}^\perp, \tag{8}$$

and, writing Π for the orthogonal projection onto $\operatorname{Ker} A \subset \mathbf{1}^\perp$,

$$\Pi C \Pi \succeq 0 \quad \text{on } \operatorname{Ker} A. \tag{9}$$

Proof. Since $q_0 = nee^*$,

$$\lambda_\varepsilon = n + \varepsilon e^* H e + O(\varepsilon^2) > 0$$

for small ε. With respect to $\mathbb{C}^n = \mathbb{C}e \oplus P\mathbb{C}^n$, the usual scalar Schur complement gives (6). Furthermore,

$$\begin{aligned} Pq_\varepsilon P &= \varepsilon PHP + \varepsilon^2 PKP + o(\varepsilon^2), \\ Pq_\varepsilon e &= \varepsilon PHe + \varepsilon^2 PKe + o(\varepsilon^2), \\ \lambda_\varepsilon^{-1} &= n^{-1} + O(\varepsilon). \end{aligned}$$

Hence

$$Pq_\varepsilon e\, \lambda_\varepsilon^{-1} e^* q_\varepsilon P = \varepsilon^2 n^{-1} PHee^*HP + o(\varepsilon^2),$$

which proves (7). If $\mathcal{F}_\varepsilon \succeq 0$, then for $v \in \mathbf{1}^\perp$,

$$0 \le v^* \mathcal{F}_\varepsilon v = \varepsilon v^* A v + \varepsilon^2 v^* C v + o(\varepsilon^2).$$

Division first by ε, and then by ε^2 on $\operatorname{Ker} A$, gives (8) and (9). □

Remark 2.2 (The curvature correction)**.** *Since* $ee^* = n^{-1}\mathbf{1}\mathbf{1}^*$,

$$C = PKP - n^{-2} PH\mathbf{1}\mathbf{1}^* HP. \tag{10}$$

The second term is the excursion from $\mathbf{1}^\perp$ *into the rank-one direction and back. Replacing* C *by* PKP *generally changes the null-sector value.*

Proposition 2.3 (Strict second-order sufficiency)**.** *Assume* $A \succeq 0$ *on* $\mathbf{1}^\perp$*. Let* Π *project onto* $\operatorname{Ker} A$ *and put* $Q = P - \Pi$*. If, for some* $\alpha, \beta > 0$*,*

$$QAQ \succeq \alpha Q, \qquad \Pi C \Pi \succeq \beta \Pi, \tag{11}$$

then $q_\varepsilon \succeq 0$ *for all sufficiently small* $\varepsilon > 0$*.*

Proof. On $Q\mathbb{C}^n$, $Q\mathcal{F}_\varepsilon Q = \varepsilon QAQ + O(\varepsilon^2)$ is positive and has inverse of order $O(\varepsilon^{-1})$. Because $A\Pi = 0 = \Pi A$, the off-diagonal block $\Pi \mathcal{F}_\varepsilon Q$ is $O(\varepsilon^2)$. The Schur complement on $\Pi\mathbb{C}^n$ is therefore

$$\varepsilon^2 \Pi C \Pi + o(\varepsilon^2) - O(\varepsilon^3),$$

which is positive for small ε by (11). Theorem 2.1 then gives $q_\varepsilon \succeq 0$. □

Remark 2.4 (Higher-order recursion requires higher-order data)**.** *The two-order hypothesis* (3) *supports only the two tests above. A recursive null-sector procedure is justified only when an asymptotic expansion through the required order, or analyticity in* ε*, is available. An* $o(\varepsilon^2)$ *remainder alone does not determine the next sign.*

3 A reflection-positive four-dimensional lattice model

3.1 Finite lattice, action, and reflections

Fix a lattice spacing $a > 0$, integers $T, L \ge 1$, and the finite reflection-symmetric lattice

$$\Lambda_{T,L} = \{-T+1, \dots, T\} \times (\mathbb{Z}/L\mathbb{Z})^3. \tag{12}$$

The first coordinate is Euclidean time. An oriented nearest-neighbour edge ℓ carries a link variable $U_\ell \in SU(2)$, with $U_{\bar{\ell}} = U_\ell^{-1}$. For a plaquette p, U_p is the ordered product of link variables around its boundary.

Let K_κ be the $SU(2)$ heat kernel at time $\kappa > 0$, in the convention

$$K_\kappa(g) = \sum_{j \in \frac{1}{2}\mathbb{N}_0} (2j+1) e^{-\kappa j(j+1)} \chi_j(g). \tag{13}$$

The finite-volume heat-kernel plaquette measure is

$$d\mu_{\Lambda,\kappa}(U) = Z_{\Lambda,\kappa}^{-1} \left(\prod_\ell dU_\ell \right) \left(\prod_p K_\kappa(U_p) \right), \tag{14}$$

where dU_ℓ is normalized Haar measure. This is a finite-dimensional probability measure. The heat-kernel action is a standard lattice Yang–Mills action and has strictly positive character coefficients [7].

Reflect the lattice through the link plane $t = \frac{1}{2}$:

$$\vartheta(t, x_1, x_2, x_3) = (1 - t, x_1, x_2, x_3). \tag{15}$$

The induced antilinear operation on observables is denoted by Θ. Let $\mathcal{A}_+$ be the gauge-invariant Wilson algebra generated by loops whose links lie entirely in the positive half $t \geq 1$.

Proposition 3.1 (Finite reflection positivity)**.** *For the measure* (14)*,*

$$\langle (\Theta F)^* F \rangle_{\Lambda,\kappa} \geq 0 \qquad (F \in \mathcal{A}_+). \tag{16}$$

Consequently every finite reflected Wilson Gram matrix is positive semidefinite.

Justification. This is the standard link-reflection positivity theorem for plaquette gauge actions with nonnegative character coefficients [6, 5]. In the heat-kernel case the coefficients in (13) are positive. The Peter–Weyl expansion factorizes each plaquette crossing the reflection plane into a sum of reflected products with nonnegative weights; plaquettes entirely on one side occur in reflected pairs. Expanding the finite product and integrating gives (16). □

We also use the spatial reflection

$$\rho(t, x_1, x_2, x_3) = (t, x_1, -x_2, x_3). \tag{17}$$

The measure is invariant under ρ, and ρ commutes with ϑ.

3.2 Three explicit closed hairpins

Let e_0, e_1, e_2, e_3 be the lattice unit vectors, with e_0 temporal. Choose positive integers r, s, h so that the following paths do not wrap around the finite box. Put

$$A = (1, 0, 0, 0), \qquad B = A + re_1,$$

and let β be the straight path from A to B. Define three detours from A to B:

$$\eta_+ : A \longrightarrow A + se_2 \longrightarrow B + se_2 \longrightarrow B, \tag{18}$$

$$\eta_- : A \longrightarrow A - se_2 \longrightarrow B - se_2 \longrightarrow B, \tag{19}$$

$$\eta_0 : A \longrightarrow A + he_0 \longrightarrow B + he_0 \longrightarrow B. \tag{20}$$

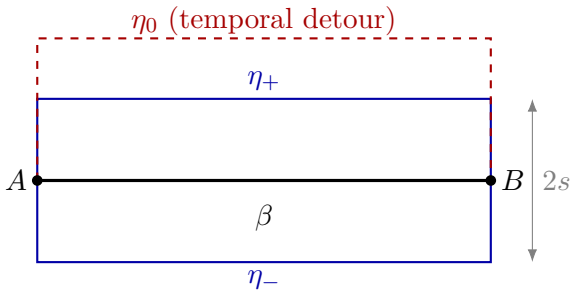


Figure 1: Schematic projection of the three detours. The dashed detour η_0 lies in the Euclidean-time direction; the picture only displays the shared-base incidence pattern.

The closed hairpin loops are

$$\gamma_\alpha = \eta_\alpha \circ \beta^{-1}, \qquad \alpha \in \{+, -, 0\}. \tag{21}$$

All links lie in $t \geq 1$. The loops share the base path β, and

$$\rho\gamma_+ = \gamma_-, \qquad \rho\gamma_0 = \gamma_0. \tag{22}$$

For a closed path γ, define the normalized fundamental Wilson loop

$$W_\gamma(U) = \frac{1}{2}\operatorname{Tr} U_\gamma. \tag{23}$$

For $SU(2)$ this is real and lies in $[-1, 1]$. Write $W_+ = W_{\gamma_+}$, $W_- = W_{\gamma_-}$, and $W_0 = W_{\gamma_0}$. Define

$$X = \frac{1}{2}(W_+ - W_-), \qquad Y = W_0 - m_0, \qquad m_0 = \langle W_0 \rangle_{\Lambda,\kappa}. \tag{24}$$

Then $X, Y \in \mathcal{A}_+$, $|X| \leq 1$, $|Y| \leq 2$, and

$$\rho X = -X, \qquad \rho Y = Y, \qquad \langle X \rangle = \langle Y \rangle = 0. \tag{25}$$

Lemma 3.2 (Symmetry decoupling)**.** *The mixed reflected correlations vanish:*

$$\langle (\Theta X)Y \rangle_{\Lambda,\kappa} = 0, \qquad \langle (\Theta Y)X \rangle_{\Lambda,\kappa} = 0. \tag{26}$$

Proof. Because the measure is ρ-invariant and ρ commutes with Θ,

$$\langle (\Theta X)Y \rangle = \langle (\Theta\rho X)(\rho Y) \rangle = -\langle (\Theta X)Y \rangle.$$

Thus the first correlation is zero. The second is its Hermitian conjugate and also vanishes. □

4 The exact three-hairpin source block

4.1 Positive-time observables and exact Gram matrix

Let

$$b = (1, -1, 0)^T, \qquad c = (0, 0, 1)^T, \qquad z = (1, 1, -2)^T. \tag{27}$$

Thus $b \perp \mathbf{1}$, $z \perp \mathbf{1}$, and $b \perp z$. For $\varepsilon \geq 0$, define three observables in the same positive-time Wilson algebra:

$$F_i^\varepsilon = 1 + \sqrt{\varepsilon}\, b_i X + \varepsilon c_i Y, \qquad i = 1, 2, 3. \tag{28}$$

The square root appears only at the observable level. The ρ parity in Lemma 3.2 removes half-integer powers from the Gram matrix.

Set

$$\sigma = \langle (\Theta X) X \rangle_{\Lambda,\kappa}, \qquad \tau = \langle (\Theta Y) Y \rangle_{\Lambda,\kappa}. \tag{29}$$

By Proposition 3.1,

$$0 \leq \sigma \leq 1, \qquad 0 \leq \tau \leq 4. \tag{30}$$

The upper bounds follow from $|X| \leq 1$ and $|Y| \leq 2$.

Proposition 4.1 (Exact reflected Gram block)**.** *The reflected Gram matrix*

$$N_\varepsilon(i,j) = \left\langle (\Theta F_i^\varepsilon)^* F_j^\varepsilon \right\rangle_{\Lambda,\kappa} \tag{31}$$

is exactly

$$N_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon\sigma b b^T + \varepsilon^2 \tau c c^T. \tag{32}$$

In particular, $N_\varepsilon \succeq 0$ for every $\varepsilon \geq 0$.

Proof. Expanding (31), the one-point terms vanish by (25). The order-$\varepsilon^{3/2}$ mixed terms vanish by (26). The remaining terms are precisely those in (32); there are no higher terms because (28) is polynomial of degree one in $\sqrt{\varepsilon}X$ and degree one in εY. Positivity also follows directly from

$$v^T N_\varepsilon v = |\mathbf{1}^T v|^2 + \varepsilon\sigma |b^T v|^2 + \varepsilon^2 \tau |c^T v|^2.$$

□

4.2 Same-theory correlation normalization

The diagonal entries of (32) are

$$N_{11} = N_{22} = 1 + \varepsilon\sigma, \qquad N_{33} = 1 + \varepsilon^2\tau. \tag{33}$$

Define

$$D_\varepsilon = \operatorname{diag}\left((1+\varepsilon\sigma)^{-1/2}, (1+\varepsilon\sigma)^{-1/2}, (1+\varepsilon^2\tau)^{-1/2} \right) \tag{34}$$

and normalized observables

$$\widetilde{F}_i^\varepsilon = (D_\varepsilon)_{ii} F_i^\varepsilon. \tag{35}$$

Their reflected Gram matrix is

$$q_\varepsilon = D_\varepsilon N_\varepsilon D_\varepsilon, \qquad (q_\varepsilon)_{ij} = \frac{N_\varepsilon(i,j)}{\sqrt{N_\varepsilon(i,i) N_\varepsilon(j,j)}}. \tag{36}$$

This is an exact observable-level congruence. It is therefore positivity preserving and does not invoke an arbitrary pair-dependent quotient.

Proposition 4.2 (Exact normalized block)**.** *The normalized matrix is*

$$q_\varepsilon = \begin{pmatrix} 1 & \alpha_\varepsilon & \beta_\varepsilon \\ \alpha_\varepsilon & 1 & \beta_\varepsilon \\ \beta_\varepsilon & \beta_\varepsilon & 1 \end{pmatrix}, \tag{37}$$

where

$$\alpha_\varepsilon = \frac{1-\varepsilon\sigma}{1+\varepsilon\sigma}, \qquad \beta_\varepsilon = \frac{1}{\sqrt{(1+\varepsilon\sigma)(1+\varepsilon^2\tau)}}. \tag{38}$$

It satisfies $q_\varepsilon \succeq 0$ for every $\varepsilon \geq 0$ and

$$q_\varepsilon \longrightarrow \mathbf{1}\mathbf{1}^T \qquad (\varepsilon \downarrow 0). \tag{39}$$

If $\sigma > 0$ and $\tau > 0$, then $q_\varepsilon \succ 0$ for every $\varepsilon > 0$, and

$$\det q_\varepsilon = \frac{4\varepsilon^3\sigma\tau}{(1+\varepsilon\sigma)^2(1+\varepsilon^2\tau)}. \tag{40}$$

Proof. Equations (37)–(38) follow from (32) and (34). Positivity and the rank-one limit follow from the congruence (36). If $\sigma, \tau > 0$ and $\varepsilon > 0$, the three linear functionals $\mathbf{1}^T v$, $b^T v$, and $c^T v$ separate points of $\mathbb{R}^3$, so (32) is positive definite. Direct calculation gives (40). □

5 Coefficient extraction and the finite certificate

Expanding (38) at $\varepsilon = 0$ gives

$$\alpha_\varepsilon = 1 - 2\sigma\varepsilon + 2\sigma^2\varepsilon^2 + O(\varepsilon^3), \tag{41}$$

$$\beta_\varepsilon = 1 - \frac{\sigma}{2}\varepsilon + \left(\frac{3\sigma^2}{8} - \frac{\tau}{2}\right)\varepsilon^2 + O(\varepsilon^3). \tag{42}$$

Thus

$$q_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon H + \varepsilon^2 K + R_\varepsilon, \tag{43}$$

with

$$H = \begin{pmatrix} 0 & -2\sigma & -\sigma/2 \\ -2\sigma & 0 & -\sigma/2 \\ -\sigma/2 & -\sigma/2 & 0 \end{pmatrix} \tag{44}$$

and

$$K = \begin{pmatrix} 0 & 2\sigma^2 & 3\sigma^2/8 - \tau/2 \\ 2\sigma^2 & 0 & 3\sigma^2/8 - \tau/2 \\ 3\sigma^2/8 - \tau/2 & 3\sigma^2/8 - \tau/2 & 0 \end{pmatrix}. \tag{45}$$

Proposition 5.1 (Uniform source-collision remainder)**.** *There is a numerical constant $C_* > 0$, independent of a, T, L, κ, r, s, h, such that*

$$\|R_\varepsilon\|_{\mathrm{op}} \leq C_*\varepsilon^3 \qquad (0 \leq \varepsilon \leq 1/4). \tag{46}$$

Proof. By (30), (σ, τ) lies in the compact set $[0,1] \times [0,4]$. The functions in (38) and their first three ε derivatives are continuous on

$$[0, 1/4] \times [0,1] \times [0,4].$$

Taylor's theorem therefore gives a uniform cubic entrywise remainder. A 3×3 Hermitian matrix has operator norm bounded by its maximum absolute row sum, which converts the entrywise bound into (46). Appendix B records an explicit admissible definition of C_*. □

Let

$$e = 3^{-1/2}\mathbf{1}, \qquad P = I - ee^T, \qquad u = 2^{-1/2}b, \qquad v = 6^{-1/2}z. \tag{47}$$

Then u, v form an orthonormal basis of $\mathbf{1}^\perp$.

Theorem 5.2 (Evaluated Schur–Feshbach gate)**.** *For the explicit block* (37)*, the first reduced coefficient is*

$$A = PHP = \sigma bb^T = 2\sigma uu^T. \tag{48}$$

The corrected second coefficient is

$$C = P\left(K - \frac{1}{3}Hee^TH\right)P = -\sigma^2 bb^T + \frac{\tau}{9}zz^T = -2\sigma^2 uu^T + \frac{2\tau}{3}vv^T. \tag{49}$$

Consequently:

(a) $A \succeq 0$ *on* $\mathbf{1}^\perp$*;*

(b) if $\sigma > 0$*, then* $\operatorname{Ker} A = \operatorname{span}\{z\}$ *on* $\mathbf{1}^\perp$*;*

(c) on that null sector,

$$z^TCz = 4\tau, \qquad v^TCv = \frac{2\tau}{3}; \tag{50}$$

(d) if $\sigma > 0$ *and* $\tau > 0$*, Proposition 2.3 applies with*

$$\alpha = 2\sigma, \qquad \beta = \frac{2\tau}{3}. \tag{51}$$

Proof. Because $b \perp \mathbf{1}$, the additive one-body terms introduced by diagonal normalization vanish after projection, and direct multiplication gives (48). Substituting (44) and (45) into Theorem 2.1 gives (49). Since $b \perp z$, the stated eigenvalues and null-sector values follow immediately. □

Remark 5.3 (The correction removes a concrete spurious term)**.** *For the null vector* z*,*

$$z^TPKPz = \sigma^2 + 4\tau, \tag{52}$$

$$z^T\left(\frac{1}{3}PHee^THP\right)z = \sigma^2, \tag{53}$$

$$z^TCz = 4\tau. \tag{54}$$

Thus the vacuum-direction coupling contributes exactly the σ^2 *term. In this example the uncorrected sign happens to remain nonnegative, but its value is wrong and would give an incorrect quantitative margin.*

Corollary 5.4 (Implemented finite Route A)**.** *For every fixed finite lattice* (12)*, every heat-kernel parameter* $\kappa > 0$*, and every admissible hairpin geometry* (18)–(21)*, the following items are established within one and the same regulated* $SU(2)$ *gauge theory:*

(i) an explicit gauge-invariant positive-time Wilson family;

(ii) the standard OS reflection used by the lattice theory;

(iii) an exact reflected Gram matrix;

(iv) a positivity-preserving normalization realized by rescaling those observables;

(v) an exact rank-one collision limit;

(vi) explicit H and K with a cutoff-uniform operator-norm remainder;

(vii) an evaluated first gate 2σ and corrected null-sector gate $2\tau/3$.

No proxy measure or pair-dependent counterterm is introduced.

6 Finite evaluation and reproducibility protocol

The only model-dependent numbers entering the closed-form block are the finite Haar integrals

$$\sigma = Z_{\Lambda,\kappa}^{-1} \int (\Theta X)(U) X(U) \prod_{\ell} dU_\ell \prod_{p} K_\kappa(U_p), \tag{55}$$

$$\tau = Z_{\Lambda,\kappa}^{-1} \int (\Theta Y)(U) Y(U) \prod_{\ell} dU_\ell \prod_{p} K_\kappa(U_p). \tag{56}$$

They can be evaluated by a finite character expansion, exact integration on a sufficiently small lattice, or reflection-symmetric Monte Carlo. Nonnegativity is already rigorous; numerical work is needed only to quantify strict margins at selected parameters.

A minimal computational record should report

$$(a, T, L, \kappa, r, s, h), \qquad \widehat{\sigma} \pm \delta_\sigma, \qquad \widehat{\tau} \pm \delta_\tau,$$

and should preserve the reflected pairing when estimating (55)–(56). A certified strict finite margin follows from

$$\widehat{\sigma} - \delta_\sigma > 0, \qquad \widehat{\tau} - \delta_\tau > 0. \tag{57}$$

The result-elaboration certificate `routea_three_hairpin_certificate.json`, generated from the supplied verification script, symbolically verifies the formulas for N_ε, q_ε, H, K, A, C, the determinant, the uncorrected null value $z^T PKPz = \sigma^2 + 4\tau$, the removed Schur–Feshbach term σ^2, and the corrected margins $u^T Au = 2\sigma$ and $v^T Cv = 2\tau/3$. This certificate supports the finite source-collision Route A block only; it is not evidence for the geometric collar modules G1–G4, endpoint modules E1–E2, or continuum/gap modules M1–M2.

7 Open modules toward a geometric collar and a mass-gap programme

The source parameter ε in (28) makes the finite block collide while the three primitive loops remain geometrically fixed. This is sufficient to instantiate the matrix theorem and validate the same-theory normalization, but it is not yet the geometric collar collapse contemplated in the original programme. The remaining steps are stated here in a form that can be filled without changing the finite result.

The following open modules have now been refined into implementation packets. Each packet separates the calculation to be performed, deterministic checks, promotion criteria, and demotion criteria. This is a manuscript integration of the expansion pipeline: it makes the next mathematical work explicit while preserving the claim boundary of the finite Route A result.

Requirement	Status in this paper	Location
Gauge group and regulator	$SU(2)$ heat-kernel plaquette measure on a finite four-dimensional lattice	Section 3
Reflection map and positive algebra	Link reflection $t \mapsto 1-t$ and Wilson algebra supported in $t \geq 1$	Proposition 3.1
Explicit Wilson geometry	Three closed hairpins sharing β	Section 3.2
Same-theory normalization	Normalized observables $\widetilde{F}_i^{\varepsilon}$; exact congruence	Equation (36)
Coefficient extraction	Closed-form H, K, and uniform $O(\varepsilon^3)$ remainder	Equations (44)–(46)
Finite test	$A = 2\sigma uu^T$, $C = -2\sigma^2 uu^T + (2\tau/3)vv^T$	Theorem 5.2
Strict finite margins	Conditional only on finite nondegeneracy $\sigma, \tau > 0$	Equation (51)
Continuum and mass gap	Not proved; separated into open modules	Section 7

Table 1: Implemented and open components of the concrete Route A realization.

7.1 Cutoff trajectory and geometric collision

A continuum-facing family would require sequences

$$a_k \downarrow 0, \qquad L_k a_k \longrightarrow \infty, \qquad T_k a_k \longrightarrow \infty, \qquad \kappa_k = \kappa(a_k), \tag{58}$$

with integer hairpin dimensions scaled to prescribed physical lengths, for example

$$a_k r_k \to R, \qquad a_k s_k \to S, \qquad a_k h_k \to H. \tag{59}$$

A genuine collar parameter δ would then enter through paths $\gamma_{i,k}(\delta)$ rather than through source amplitudes alone.

Open module G1: geometric family. This module is not yet a theorem. It is implemented as a precise construction target for a cutoff-indexed geometric collar.

Target. construct paths $\gamma_{+,k}(\delta), \gamma_{-,k}(\delta), \gamma_{0,k}(\delta)$ with locked reflected boundary footprint.

Needed calculations.

(a) Choose integers $r_k, s_k, h_k, m_k(\delta)$ with $a_k r_k \to R$, $a_k s_k \to S$, $a_k h_k \to H$, and $a_k m_k(\delta) \to \delta$.

(b) Write each path as an ordered edge list using A_k, $B_k = A_k + r_k e_1$, transverse offsets $\pm m_k(\delta) e_2$, and the temporal offset $h_k e_0$.

(c) Check every edge has time coordinate ≥ 1, or move the base time from 1 to $1 + h_k$ and record the shift.

(d) Compute the common reflected boundary footprint and prove it is independent of the collar sign $\pm\delta$.

(e) Prove $\rho\gamma_{+,k}(\delta) = \gamma_{-,k}(\delta)$ and $\rho\gamma_{0,k}(\delta) = \gamma_{0,k}(\delta)$.

Promotion criterion. Promote G1 only when the edge-list construction satisfies positive-time support, footprint locking, reflection pairing, and the stated joint scaling.

Demotion boundary. Keep G1 open if the collar collapse changes the footprint, exits the positive algebra, or depends on an unstated order of limits.

Open module G2: renormalized same-theory normalization. This module is implemented as a normalization audit and theorem target, not as a completed renormalization proof.

Target. prove that all renormalized geometric-collar normalizations are same-theory positive operations.

Needed calculations.

(a) List every normalization factor for $N_{k,\delta}$: perimeter, cusp, boundary, endpoint, and overlap factors.

(b) Classify each factor as one-body $a_i\overline{a}_j$, block congruence, pair-dependent B_{ij}, or positive-kernel candidate.

(c) For one-body factors define $D_{k,\delta}$ and verify $q_{k,\delta} = D_{k,\delta}N_{k,\delta}D^*_{k,\delta}$.

(d) For pair-dependent factors test whether B^{-1} is positive definite or can be represented as a Schur product with a positive kernel.

(e) Record any factor that fails these tests as a demotion blocker for the geometric-collar application.

Promotion criterion. Promote G2 only when every removed factor is realized by congruence, direct normalized observables, or a proved positive-kernel factorization.

Demotion boundary. Demote the geometric route if it needs pair-dependent entrywise division without a positivity certificate.

Open module G3: geometric coefficient extraction. This module is implemented as the exact coefficient-extraction workload for the geometric collar; no geometric H,K theorem is claimed yet.

Target. derive $q_{k,\delta} = \mathbf{1}\mathbf{1}^T + \delta H_k + \delta^2 K_k + R_{k,\delta}$ from the geometric collar.

Needed calculations.

(a) Express every entry of the normalized geometric block $q_{k,\delta}(i,j)$ as a same-theory Wilson expectation.

(b) Choose whether to expand entries directly or expand $\log q_{k,\delta}(i,j)$, then state the conversion to H_k, K_k.

(c) Compute first variations in δ and isolate additive one-body terms killed by P.

(d) Compute second variations and separate genuine K_k terms from Schur correction terms.

(e) Prove $\sup_k \|R_{k,\delta}\|/\delta^2 \to 0$ in the declared a_k, δ regime.

Promotion criterion. Promote G3 only when H_k, K_k and the operator-norm remainder are derived from the same normalized geometric block.

Demotion boundary. Do not promote the source-amplitude H, K calculation as a geometric-collar coefficient theorem.

Open module G4: uniform nondegeneracy. This module is implemented as a lower-bound programme for the reduced margins, not as a proved continuum lower bound.

Target. prove or certify lower bounds for the renormalized analogues of σ_k and τ_k.

Needed calculations.

(a) Define the renormalized $\sigma_k(\delta)$ and $\tau_k(\delta)$ after G2 normalization.

(b) Separate positivity from strict positivity: reflection positivity gives ≥ 0, not a lower bound.

(c) Identify whether lower bounds come from symmetry, transfer-matrix estimates, character expansion, or certified finite computation.

(d) Propagate certified intervals into $2\sigma_k$ and $2\tau_k/3$.

(e) State the limiting margin condition needed before revising the abstract toward geometric-collar success.

Promotion criterion. Promote G4 only when lower bounds survive the chosen cutoff trajectory.

Demotion boundary. Keep strict-margin claims finite if σ_k or τ_k may vanish as $k \to \infty$.

7.2 Correct endpoint/null-channel bookkeeping

Let R be a real symmetric coefficient matrix in a normalized 3×3 block and let $z = (1, 1, -2)^T$. The exact identity is

$$z^T R z = R_{11} + R_{22} + 4R_{33} + 2R_{12} - 4R_{13} - 4R_{23}. \tag{60}$$

Only when the normalization enforces $R_{11} = R_{22} = R_{33} = 0$ does this reduce to the incidence weights

$$z^T R z = 2R_{12} - 4R_{13} - 4R_{23}. \tag{61}$$

The exact block (37) has unit diagonal for every ε, so every coefficient beyond the leading term has zero diagonal and (61) applies.

Suppose an endpoint calculation produces a computable approximation $\widehat{R}_{\mathrm{end}}$ and a certified two-sided error

$$|R_{\mathrm{end}} - \widehat{R}_{\mathrm{end}}| \leq \Delta_{\mathrm{end}}. \tag{62}$$

Lemma 7.1 (Logically valid endpoint certificates)**.** *Let M_G be the finite positive margin before the endpoint correction. Under (62),*

$$M_G + \widehat{R}_{\mathrm{end}} - \Delta_{\mathrm{end}} > 0 \quad \Longrightarrow \quad M_G + R_{\mathrm{end}} > 0, \tag{63}$$

$$M_G + \widehat{R}_{\mathrm{end}} + \Delta_{\mathrm{end}} < 0 \quad \Longrightarrow \quad M_G + R_{\mathrm{end}} < 0. \tag{64}$$

If one has only a lower bound $R_{\mathrm{end}} \geq -U$, then $M_G - U > 0$ is a sufficient positive certificate, while $U > M_G$ is inconclusive.

Proof. Equation (62) gives

$$\widehat{R}_{\mathrm{end}} - \Delta_{\mathrm{end}} \le R_{\mathrm{end}} \le \widehat{R}_{\mathrm{end}} + \Delta_{\mathrm{end}}.$$

Adding M_G proves both implications. A one-sided lower bound cannot certify negativity. □

A heat-kernel endpoint implementation could seek an estimator of the schematic form

$$\widehat{R}_{\mathrm{end}} = C_{\mathrm{col}} \sum_v \left(2\widehat{\kappa}^{12}_{4,v} - 4\widehat{\kappa}^{13}_{4,v} - 4\widehat{\kappa}^{23}_{4,v} \right) + \widehat{E}_{\mathrm{tail}}, \tag{65}$$

with a two-sided error obtained from source-derivative and tail estimates.

Open module E1: endpoint regulator derivation. This module is implemented as a same-regulator endpoint derivation target, not as an endpoint theorem.

Target. derive the endpoint estimator and color/incidence factors from the SU(2) heat-kernel regulator.

Needed calculations.

(a) Define endpoint boxes from the G1 geometric paths, including which plaquettes and links belong to each box.

(b) Define source variables and singlet projections in the fundamental $SU(2)$ representation.

(c) Derive C_{col} from the heat-kernel group integral with stated trace normalization.

(d) Derive the incidence weights in $2\widehat{\kappa}^{12}_{4,v} - 4\widehat{\kappa}^{13}_{4,v} - 4\widehat{\kappa}^{23}_{4,v}$ from path geometry, not by assumption.

(e) Check diagonal terms vanish only under the unit-diagonal normalization condition.

Promotion criterion. Promote E1 only when the estimator is derived from the same geometric Wilson family and regulator.

Demotion boundary. Do not infer endpoint signs from an incidence ansatz not tied to the regulated path integral.

Open module E2: cumulant and tail bounds. This module is implemented as the two-sided error architecture needed for endpoint certificates.

Target. prove $|R_{\mathrm{end}} - \widehat{R}_{\mathrm{end}}| \le \Delta_{\mathrm{end}}$.

Needed calculations.

(a) State the compact source domain for every endpoint logarithm $\ell_v(s) = \log Z_v(s)$.

(b) Prove or bound $\sup |\ell_v^{(6)}(s)|$ to justify the five-point fourth-derivative stencil.

(c) Bound endpoint, corner, inter-box, and tail terms on both sides.

(d) Propagate the two-sided error into $M_G + \widehat{R}_{\mathrm{end}} \pm \Delta_{\mathrm{end}}$.

(e) Classify the result as positive certificate, negative certificate, or inconclusive; never use a one-sided lower bound as a negative certificate.

Promotion criterion. Promote E2 only with a same-regulator two-sided error bound.

Demotion boundary. A one-sided lower bound can certify positivity when $M_G - U > 0$, but cannot certify an obstruction.

7.3 From OS positivity to a spectral gap

Reflection positivity of finite blocks is necessary for OS reconstruction but does not imply a spectral gap. A mass-gap argument must additionally produce the continuum Schwinger functions, the reconstructed Hilbert space and Hamiltonian, and a uniform decay or spectral estimate in that same theory.

Open module M1: continuum reconstruction. This module is implemented as a reconstruction handoff checklist; it remains open.

Target. construct continuum Schwinger functions satisfying the full OS axioms on a separating Wilson algebra.

Needed calculations.

(a) List the OS axioms still missing: Euclidean invariance, symmetry, regularity, clustering/nontriviality as applicable, and reflection positivity for a separating algebra.

(b) Specify the topology for $a_k \to 0$, $L_k a_k, T_k a_k \to \infty$.

(c) State tightness or compactness estimates needed for Wilson-loop distributions.

(d) Identify how finite geometric-collar blocks embed into the continuum observable algebra.

(e) Name which result would produce the reconstructed Hilbert space and Hamiltonian.

Promotion criterion. Promote M1 only with an actual continuum construction or a theorem importing one in the same theory.

Demotion boundary. Finite reflection positivity must remain a necessary finite condition, not OS reconstruction.

Open module M2: gap estimate. This module is implemented as a spectral-gap handoff checklist; it remains open.

Target. derive a same-theory estimate implying no Hamiltonian spectrum in $(0, \Delta)$.

Needed calculations.

(a) Define the reconstructed Hamiltonian H_{YM}, vacuum Ω, and excitation sector.

(b) State the Euclidean decay, transfer-matrix, or correlation estimate sufficient for a spectral gap.

(c) Check that the estimate is in the same continuum theory produced by M1.

(d) Audit every step for circular use of exponential decay, clustering, or mass gap.

(e) State what observable sector witnesses the first non-vacuum excitation.

Promotion criterion. Promote M2 only after M1 and a non-circular same-theory spectral estimate.

Demotion boundary. No finite-block or endpoint result may be described as a mass-gap proof.

8 Conclusion

The finite Schur–Feshbach coefficient test has been implemented in a concrete reflection-positive lattice gauge theory rather than left as a conditional Wilson template. The construction fixes the compact simple group $SU(2)$, a four-dimensional heat-kernel plaquette action, the time reflection, three closed hairpin loops, a symmetry-adapted positive-time Wilson family, and an exact observable-level normalization. The resulting block is explicitly calculable and positive at every finite cutoff.

The main quantitative output is

$$A = 2\sigma uu^T, \qquad C = -2\sigma^2 uu^T + \frac{2\tau}{3} vv^T.$$

The first reduced margin is 2σ and the corrected null-sector margin is $2\tau/3$. The calculation also exhibits the purpose of the Schur correction: it removes the spurious σ^2 contribution from the second null-channel value.

What remains is now sharply separated. The verified calculation is the finite source-collision algebra: it certifies the displayed 3×3 block, the cancellation of the spurious σ^2 null-channel term, and the reduced margins. The present source collision must still be replaced or supplemented by a lattice-resolved geometric collar; renormalized coefficients and strict margins must be controlled along a continuum trajectory; endpoint cumulants need two-sided, regulator-specific error estimates; and reflection positivity must be joined to continuum reconstruction and a spectral estimate. Those are open modules, not consequences of the finite block or of the verification certificate.

A Normalization algebra

Lemma A.1 (One-body normalization is a congruence)**.** *Let $N = N^*$ and let $D = \mathrm{diag}(d_1, \dots, d_n)$ be invertible. Then*

$$N \succeq 0 \quad \Longleftrightarrow \quad DND^* \succeq 0.$$

Proof. For every x, $x^* DND^* x = (D^* x)^* N (D^* x)$, and D^* is bijective. □

In particular, the correlation normalization

$$q_{ij} = \frac{N_{ij}}{\sqrt{N_{ii} N_{jj}}}$$

is a congruence whenever $N_{ii} > 0$. It is realized by the normalized observables $F_i/\sqrt{N_{ii}}$.

Lemma A.2 (An uncontrolled entrywise quotient need not preserve positivity)**.** *There are positive-semidefinite matrices N and B, with every entry of B positive, such that $N \oslash B$ is indefinite.*

Proof. Take

$$N = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1/10 \\ 1/10 & 1 \end{pmatrix}.$$

Both are positive semidefinite, whereas

$$N \oslash B = \begin{pmatrix} 1 & 10 \\ 10 & 1 \end{pmatrix}$$

has eigenvalues 11 and -9. □

Lemma A.3 (Zero-sum cancellation of additive one-body terms)**.** *For $d \in \mathbb{C}^n$, let*

$$L = \frac{1}{2}(d\mathbf{1}^* + \mathbf{1}d^*).$$

Then $PLP = 0$.

Proof. For $v, w \in \mathbf{1}^\perp$, both $\mathbf{1}^* w$ and $v^*\mathbf{1}$ vanish, so $v^* L w = 0$. □

B Exact block algebra and uniform remainder

From (32),

$$N_\varepsilon = \begin{pmatrix} 1+\varepsilon\sigma & 1-\varepsilon\sigma & 1 \\ 1-\varepsilon\sigma & 1+\varepsilon\sigma & 1 \\ 1 & 1 & 1+\varepsilon^2\tau \end{pmatrix}.$$

Applying (34) gives (37). For a fully explicit uniform Taylor constant, define

$$f(\varepsilon,\sigma) = \frac{1-\varepsilon\sigma}{1+\varepsilon\sigma},$$
$$g(\varepsilon,\sigma,\tau) = \big((1+\varepsilon\sigma)(1+\varepsilon^2\tau)\big)^{-1/2}$$

and

$$M_* = \max_{\substack{0\le t\le 1/4 \\ 0\le\sigma\le 1 \\ 0\le\tau\le 4}} \max\left\{|\partial_t^3 f(t,\sigma)|, |\partial_t^3 g(t,\sigma,\tau)|\right\}. \tag{66}$$

The maximum is finite by compactness and positivity of the denominators. Taylor's theorem gives entrywise remainders at most $M_*\varepsilon^3/6$. The remainder matrix has zero diagonal and off-diagonal pattern (r_{12}, r_{13}, r_{13}), so its maximum row sum is at most $M_*\varepsilon^3/3$. Thus one may take

$$C_* = M_*/3 \tag{67}$$

in Proposition 5.1.

For the Schur coefficients, direct multiplication gives

$$PHP = \sigma bb^T,$$
$$P\left(K - \frac{1}{3}Hee^TH\right)P = -\sigma^2 bb^T + \frac{\tau}{9}zz^T.$$

Since $b^Tb = 2$, $z^Tz = 6$, and $b^Tz = 0$, the reduced eigenvalues are 2σ and $2\tau/3$.

The determinant can also be obtained before normalization. Since

$$\det N_\varepsilon = 4\varepsilon^3\sigma\tau$$

and

$$\det D_\varepsilon^2 = \frac{1}{(1+\varepsilon\sigma)^2(1+\varepsilon^2\tau)},$$

Equation (40) follows.

C Five-point fourth-derivative certificate

Lemma C.1 (Central five-point certificate)**.** *Let $f \in C^6([-2h, 2h])$ and define*

$$D_h f = \frac{f(2h) - 4f(h) + 6f(0) - 4f(-h) + f(-2h)}{h^4}. \tag{68}$$

Then

$$\left| D_h f - f^{(4)}(0) \right| \le \frac{h^2}{6} \sup_{|s| \le 2h} |f^{(6)}(s)|. \tag{69}$$

Proof. It is enough to prove the unit-mesh result and apply it to $g(u) = f(hu)$. The functional

$$L(g) = g(2) - 4g(1) + 6g(0) - 4g(-1) + g(-2) - g^{(4)}(0)$$

annihilates polynomials of degree at most five. Peano's kernel theorem gives

$$L(g) = \int_{-2}^{2} K(t) g^{(6)}(t)\, dt,$$

where the kernel is nonnegative and has integral $1/6$. Hence

$$|L(g)| \le \frac{1}{6} \sup_{|t| \le 2} |g^{(6)}(t)|.$$

Since $g^{(4)}(0) = h^4 f^{(4)}(0)$ and $g^{(6)}(u) = h^6 f^{(6)}(hu)$, division by h^4 proves (69). □

For an endpoint source logarithm $\ell_v(s) = \log Z_v(s)$, the lemma turns a formal fourth cumulant into a finite-difference estimator. A usable certificate still requires a uniform bound on $\ell_v^{(6)}$ and a two-sided estimate of all tails, as stated in Open module E2.

D Checklist for the next manuscript revision

P1. Supply affiliation, contact information, acknowledgements, and a data/code statement.

P2. Choose at least one numerical finite lattice (T, L, κ, r, s, h) and evaluate σ and τ with certified uncertainties.

P3. Decide whether the principal next theorem concerns a source collision, a geometric collar collision, or both; keep the parameters notationally distinct.

P4. For a geometric collar, provide path diagrams at every cutoff and specify the joint order of $a \downarrow 0$ and $\delta \downarrow 0$.

P5. Derive any endpoint cumulant representation from the chosen action and path incidence; do not infer a negative obstruction from failure of a one-sided positive lower bound.

P6. Add a continuum theorem only after the required uniform estimates are proved. Keep the mass-gap claim separate from finite reflection positivity.

References

[1] C. N. Yang and R. L. Mills. Conservation of isotopic spin and isotopic gauge invariance. *Physical Review* 96, 191–195, 1954.

[2] K. Osterwalder and R. Schrader. Axioms for Euclidean Green's functions. *Communications in Mathematical Physics* 31, 83–112, 1973. doi:10.1007/BF01645738.

[3] K. Osterwalder and R. Schrader. Axioms for Euclidean Green's functions II. *Communications in Mathematical Physics* 42, 281–305, 1975. doi:10.1007/BF01608978.

[4] K. G. Wilson. Confinement of quarks. *Physical Review D* 10, 2445–2459, 1974. doi:10.1103/PhysRevD.10.2445.

[5] M. Lüscher. Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories. *Communications in Mathematical Physics* 54, 283–292, 1977. doi:10.1007/BF01614090.

[6] K. Osterwalder and E. Seiler. Gauge field theories on a lattice. *Annals of Physics* 110, 440–471, 1978.

[7] P. Menotti and E. Onofri. The action of $SU(N)$ lattice gauge theory in terms of the heat kernel on the group manifold. *Nuclear Physics B* 190, 288–300, 1981.

[8] A. Shapiro. First and second order analysis of nonlinear semidefinite programs. *Mathematical Programming* 77, 301–320, 1997.

[9] J. F. Bonnans and A. Shapiro. *Perturbation Analysis of Optimization Problems*. Springer, New York, 2000.

[10] A. Jaffe and E. Witten. Quantum Yang–Mills theory. Clay Mathematics Institute Millennium Problem statement, 2000. https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf

[11] Clay Mathematics Institute. Yang–Mills and the Mass Gap. https://www.claymath.org/millennium/yang-mills-the-maths-gap/